IRRIGATION WATER MANAGEMENT · NOTE WM01

Measuring Irrigation Water — Units, Volume and Velocity–Area Methods, Water Meters

Course topics: IDE5.5 L6, CAE1.3 L11, IDE3.6 L12, IDE8.17 L8, IDE-509 T9

Why measure irrigation water

- Water is the most valuable asset of irrigated farming. Measuring it accurately leads to economical, efficient use; efficient conveyance and proper distribution raise irrigation efficiency and crop productivity.
- Measurement lets you apply the correct quantity, match application rates to the soil's infiltration, cut waste and prevent land damage.
- It is essential for soil–water–plant studies and for charging farmers by the actual volume supplied.
- Knowing soil–moisture–plant relations lets engineers design systems that apply the right amount — delivering water in proper amounts cuts waste and improves water-use efficiency.

Units

Water is measured at rest (as a volume) or in motion (as a rate of flow — the volume passing a section per unit time).

Unit	Definition and use
Litre	One cubic decimetre, or 1/1000 cubic metre
Cubic metre	Volume of a 1 m × 1 m × 1 m cube = 1000 litres
Hectare-centimetre	Water to cover 1 hectare (10,000 m²) to a depth of 1 centimetre
Litre per second (L/s)	A continuous flow of 1 litre past a point each second — for pumps, pipelines and small streams
Cubic metre per second (m³/s)	A continuous flow of 1 cubic metre (1000 litres) per second — for canals, streams and rivers
Litre per minute (L/min)	Small flows

Volume method

1. Collect the discharge in a container of known volume — an ordinary bucket, or a barrel of about 150–200 litres or more for pumps.
2. Time how long it takes to fill, with a stopwatch or the second hand of a wristwatch.
3. Compute the discharge.

$$\text{Discharge (L/s)} = \text{volume of container (litres)} \;/\; \text{time to fill (seconds)}$$

Use it for pumps and for manual and animal-powered water lifts.

Velocity–area method (streams)

$$\text{Discharge} = \text{velocity} \times \text{area of cross-section}$$

Float method (approximate)

1. Choose a straight reach about 30 m long with a fairly uniform section; measure the depth and width at several points.
2. Stretch a string across each end at right angles to the flow.
3. Release a float upstream and time it over the reach; repeat several times and average.
4. Velocity = length of reach / time of travel. The surface moves faster than the stream as a whole, so multiply by a coefficient, usually 0.85.

$$\text{Average velocity} = \text{measured velocity} \times 0.85;\ \text{Discharge} = \text{average velocity} \times \text{average cross-sectional area}$$

Use it where costly installations are not justified and high accuracy is not needed.

Current-meter method

- A current meter has a wheel or vane that the flowing water turns; hang it on a cable in deep streams or fix it to a rod in shallow ones. Count the revolutions over a time and read the velocity from its calibration table or graph.
- Divide the section into sub-areas 0.6 m to 6 m wide (depending on stream size and precision) and gauge each separately. Estimate each mean velocity from readings at 0.2 and 0.8 of the depth below the surface.
- Discharge = mean velocity × area of cross-section (summed over the sub-areas).

Both float and current-meter methods have very limited use in farm irrigation practice.

Water meters (in pipelines)

- Propeller-type meters are most common: a register on top shows the total volume passed, and some models show the instantaneous rate. Keep debris out of the propeller.
- For accuracy, the pipe must flow full at all times and the flow must exceed the meter's minimum rated flow.
- Meters are factory-calibrated and seldom need field adjustment. In an open channel, bring the flow through a pipe of known cross-section to meter it.

Sources: ICAR eCourse 'Irrigation Engineering' — Dr K. N. Tiwari and Dr N. S. Raghuwanshi (IIT Kharagpur), reviewed by Dr S. D. Gorantiwar (MPKV Rahuri). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. One hectare-centimetre of water equals

(a) 10 m³ (b) 100 m³
(c) 1000 m³ (d) 10,000 m³

Q2. In the float method, surface velocity is multiplied by about

(a) 0.5 (b) 0.65
(c) 0.85 (d) 1.15

Q3. With a current meter, mean velocity is estimated from readings at

(a) 0.1 and 0.9 depth
(b) 0.2 and 0.8 depth
(c) 0.5 depth only
(d) surface only

Q4. A water meter in a pipeline reads accurately only when the pipe

(a) flows half full
(b) flows full
(c) is empty
(d) carries air

Q5. A 200 L barrel fills in 25 s. The pump discharge is

(a) 5 L/s
(b) 8 L/s
(c) 12.5 L/s
(d) 25 L/s

Answers

Q1 — (b) 100 m³ — *10,000 m² × 0.01 m.*
Q2 — (c) 0.85
Q3 — (b) 0.2 and 0.8 depth
Q4 — (b) flows full
Q5 — (b) 8 L/s — *200/25 = 8 L/s.*